

Credit Card Approval Prediction

Project Description:

The Credit Card Approval Prediction System is a Machine Learning-based web application that automates the process of evaluating credit card applications. Financial institutions receive thousands of applications every day, making manual verification slow, expensive, and susceptible to human errors. This project aims to simplify and accelerate the approval process by predicting whether a credit card application is likely to be approved or rejected based on the applicant's financial and demographic information.

The system utilizes four supervised machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Historical credit card applicant data is preprocessed, cleaned, and used to train these models. Their performances are compared using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The best-performing model is saved using Pickle and integrated into a Flask web application for real-time predictions.

To make the application scalable and cloud-accessible, the trained model is deployed using IBM Watson Machine Learning. Users can access the prediction system through an intuitive web interface where they enter applicant details and instantly receive an approval prediction.

Scenarios

Scenario 1: Automated Credit Card Application Screening

A bank employee enters an applicant's details such as annual income, employment status, loan amount, credit history, and existing debts into the web application. The machine learning model instantly predicts whether the application should be approved or rejected, reducing manual workload and improving decision-making speed.

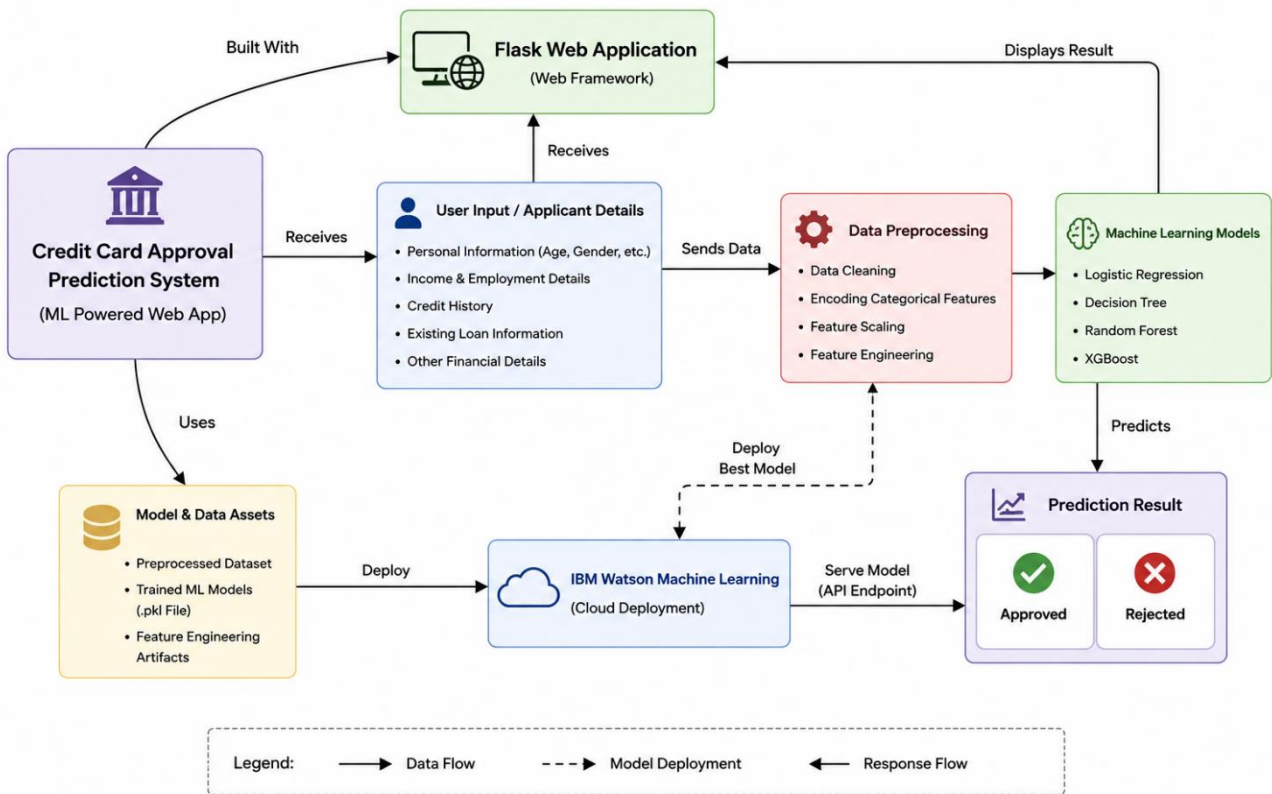
Scenario 2: High-Risk Applicant Identification

A compliance officer screens applicants with previous loan defaults and poor repayment records. The system identifies applicants with high financial risk and recommends rejection, helping banks reduce loan defaults and financial losses.

Scenario 3: Customer Self-Service Eligibility Check

A customer visits the web portal before applying for a credit card. After entering financial information such as income, employment status, and credit history, the system predicts the likelihood of approval, allowing customers to understand their eligibility before submitting an official application.

Technical Architecture:



Description: The Credit Card Approval Prediction System is built using the Flask web framework, which provides a user-friendly interface for entering applicant details. The application preprocesses the input data and sends it to trained machine learning models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost. The best-performing model predicts whether the credit card application will be approved or rejected based on the applicant's financial profile. The trained model is deployed using IBM Watson Machine Learning to enable secure and scalable cloud-based predictions. Finally, the prediction result is displayed instantly to the user through the Flask web application.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

sPre-requisites:

- **Python Programming Proficiency:** Python Documentation
- **Machine Learning Fundamentals:** Scikit-learn Documentation
- **Flask Framework Knowledge:** Flask Documentation
- **Pandas and NumPy for Data Processing:** Pandas & NumPy Documentation
- **XGBoost Library Familiarity:** XGBoost Documentation
- **HTML, CSS, and JavaScript Skills:** W3Schools HTML/CSS/JavaScript Tutorials
- **IBM Watson Machine Learning Knowledge:** IBM Cloud & Watson Machine Learning Documentation
- **Model Serialization using Pickle:** Python Pickle Documentation
- **Version Control with Git:** Git Documentation
- **Development Environment Setup:** Flask Installation Guide

Dataset Requirements

- Historical Credit Card Approval Dataset
- Applicant Financial Information
- Credit History Records
- Income and Employment Details

Python Libraries

- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Flask
- Joblib/Pickle
- Matplotlib

Technical Knowledge

- Python Programming
- Machine Learning Basics
- Data Preprocessing Techniques
- Classification Algorithms
- Flask Web Development
- HTML, CSS, and JavaScript
- REST API Basics
- Version Control using Git

Project Workflow: Milestone 1: Registration Page



Create an account
Register to predict credit card application decisions

USERNAME

Alex

EMAIL ADDRESS

Alex@gmail.com

PASSWORD

CONFIRM PASSWORD

Create Account

Already have an account? [Log in](#)

Milestone 2: Login Page



Welcome back
Log in to manage credit card applications

EMAIL ADDRESS

you@example.com

PASSWORD

Sign In

Don't have an account? [Create an account](#)

Milestone 3 : User Profile


CreditShield



Alex
 Alex@gmail.com

AL

Logout

NAVIGATION MENU

- Dashboard
- Apply / Predict
- Prediction History
- User Profile**

Account Settings

Manage your login credentials and profile details.



Alex
 Alex@gmail.com
 • Verified Officer

USERNAME

EMAIL ADDRESS

NEW PASSWORD (OPTIONAL)

CONFIRM PASSWORD

Update Profile

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Milestone 4: Application Page


CreditShield



Alex
 Alex@gmail.com

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NAVIGATION MENU

- Dashboard
- Apply / Predict**
- Prediction History
- User Profile

Apply / Predict Approval

Run credit evaluation algorithm using customer demographic and credit profile.

GENDER

AGE

ANNUAL INCOME (\$)

EMPLOYMENT TYPE

EDUCATION

MARITAL STATUS

DEPENDENTS

CREDIT SCORE (300 - 850)

REQUESTED LIMIT / LOAN (\$)

EXISTING LOAN

PROPERTY AREA

CREDIT HISTORY



Evaluation Pending

Fill in the customer parameters on the left and click "Predict Approval" to see the credit score check outcome.

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Milestone 5: Evaluation Page(Approved)


CreditShield


 Alex
 Alex@gmail.com
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NAVIGATION MENU

- Dashboard
- Apply / Predict**
- Prediction History
- User Profile

GENDER Female	AGE 33
ANNUAL INCOME (\$) 50000	EMPLOYMENT TYPE Employed
EDUCATION Graduate	MARITAL STATUS Married
DEPENDENTS 2	CREDIT SCORE (300 - 850) 500
REQUESTED LIMIT / LOAN (\$) 50000	EXISTING LOAN Yes
PROPERTY AREA Urban	CREDIT HISTORY Good (No defaults)

Reset Fields
 Predict Approval



APPLICATION APPROVED

MODEL CONFIDENCE

60.4%

DECISION REASONING

Good credit history indicates reliable repayment behavior. Stable employment type reduces income default risks.

Milestone 6: Evaluation Page(Rejected)


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NAVIGATION MENU

- Dashboard
- Apply / Predict**
- Prediction History
- User Profile

GENDER Male	AGE 44
ANNUAL INCOME (\$) 30000	EMPLOYMENT TYPE Unemployed
EDUCATION Not Graduate	MARITAL STATUS Divorced
DEPENDENTS 2	CREDIT SCORE (300 - 850) 300
REQUESTED LIMIT / LOAN (\$) 300000	EXISTING LOAN Yes
PROPERTY AREA Urban	CREDIT HISTORY Good (No defaults)

Reset Fields
 Predict Approval



APPLICATION REJECTED

MODEL CONFIDENCE

55.2%

DECISION REASONING

Low credit score of 300 is below the acceptable threshold. High requested loan amount (\$300,000) relative to income (\$30,000) increases debt burden (DTI: 1000.0%). Lack of steady employment increases default risk. Multiple active lines of credit combined with average credit score limit borrowing capacity.

Milestone 7: Prediction History Page

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NAVIGATION MENU

 Dashboard

 Apply / Predict

 Prediction History

 User Profile

Prediction History

Audit history of past customer queries and machine learning prediction outcomes.

STATUS: All Predictions

Export CSV

Clear History

DATE	GENDER	AGE	INCOME	CREDIT SCORE	LOAN AMOUNT	CREDIT HISTORY	PREDICTION	CONFIDENCE
2026-07-10	Male	44	\$30,000	300	\$300,000	Good	Rejected	55.2%
2026-07-10	Female	33	\$50,000	500	\$50,000	Good	Approved	60.4%

Showing 1 to 2 of 2 entries

Previous

Page 1 of 1

Next

Conclusion:

The Credit Card Approval Prediction System successfully automates the process of evaluating credit card applications using machine learning techniques. By comparing multiple classification algorithms—Logistic Regression, Decision Tree, Random Forest, and XGBoost—the project identifies the best-performing model to accurately predict whether an application should be approved or rejected. The trained model is integrated into a Flask web application, providing users with instant and reliable predictions through a simple interface. Furthermore, deployment using IBM Watson Machine Learning enables secure, scalable, and cloud-based access to the prediction service. This project helps financial institutions reduce manual effort, improve decision-making accuracy, and speed up the credit approval process, while offering a strong foundation for future enhancements such as explainable AI, fraud detection, and real-time banking system integration.